MSG No. 140.549 $I4/m'c'm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `422`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'2m'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,5,6,26,27,31,32]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,7,8,25,28,29,30]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,12,13,14,18,19,23,24]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[10,11,15,16,17,20,21,22]`

Table 3: Wyckoff site: `4c`, site symmetry: `4/m'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,6,9,10,11,14]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,7,8,12,13,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,18,19,22,25,26,27,30]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[20,21,23,24,28,29,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `m'.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,6,9,14,23,24,31,32]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,3,10,11,20,21,28,29]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,5,12,13,18,19,26,27]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[7,8,15,16,17,22,25,30]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,7,25,31]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,5,26,29]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,4,27,28]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[6,8,30,32]`
5	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[9,15,17,23]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[10,13,18,21]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[11,12,19,20]`

8	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[14,16,22,24]`

Table 6: Wyckoff site: `8f`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,7,8]`
3	$[0,\ 0,\ -z]$	`[9,10,11,14]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[12,13,15,16]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,18,19,22]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[20,21,23,24]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[25,26,27,30]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[28,29,31,32]`

Table 7: Wyckoff site: `8g`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,31,32]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,28,29]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4,5,26,27]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[7,8,25,30]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,14,23,24]`
6	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[10,11,20,21]`
7	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,13,18,19]`
8	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[15,16,17,22]`

Table 8: Wyckoff site: `8h`, site symmetry: `m'.2m'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,14,23,32]`
2	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[2,11,21,28]`
3	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[3,10,20,29]`
4	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[4,13,19,26]`
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[5,12,18,27]`
6	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[6,9,24,31]`
7	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[7,16,17,30]`
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	`[8,15,22,25]`

Table 9: Wyckoff site: `16i`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,7]`
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	`[2,5]`
3	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[3,4]`

continued ...

Table 9

No.	position	mapping
4	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[6,8]
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[9,15]
6	$\left[x,\ -x,\ \frac{3}{4}\right]$	[10,13]
7	$\left[-x,\ x,\ \frac{3}{4}\right]$	[11,12]
8	$\left[x,\ x,\ \frac{3}{4}\right]$	[14,16]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[17,23]
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[18,21]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[19,20]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[22,24]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[25,31]
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[26,29]
15	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[27,28]
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[30,32]

Table 10: Wyckoff site: 16j, site symmetry: .2.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,4]
2	$\left[0,\ x,\ \frac{1}{4}\right]$	[2,7]
3	$\left[0,\ -x,\ \frac{1}{4}\right]$	[3,8]
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[5,6]
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[9,12]
6	$\left[0,\ -x,\ \frac{3}{4}\right]$	[10,15]
7	$\left[0,\ x,\ \frac{3}{4}\right]$	[11,16]
8	$\left[x,\ 0,\ \frac{3}{4}\right]$	[13,14]
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[17,20]
10	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[18,23]
11	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[19,24]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[21,22]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[25,28]
14	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[26,31]
15	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[27,32]
16	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[29,30]

Table 11: Wyckoff site: 16k, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$[-y,\ x,\ 0]$	[2,11]
3	$[y,\ -x,\ 0]$	[3,10]
4	$\left[x,\ -y,\ \frac{1}{2}\right]$	[4,13]
5	$\left[-x,\ y,\ \frac{1}{2}\right]$	[5,12]
6	$[-x,\ -y,\ 0]$	[6,9]
7	$\left[y,\ x,\ \frac{1}{2}\right]$	[7,16]

continued ...

Table 11

No.	position	mapping
8	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[8,15]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[17,30]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,27]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[19,26]
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[20,29]
13	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[21,28]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[22,25]
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[23,32]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	[24,31]

Table 12: Wyckoff site: 16l, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	[1,32]
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[2,28]
3	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[3,29]
4	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[4,26]
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[5,27]
6	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[6,31]
7	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[7,30]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[8,25]
9	$\left[-x,\ \frac{1}{2}-x,\ -z\right]$	[9,24]
10	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[10,20]
11	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[11,21]
12	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[12,18]
13	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[13,19]
14	$\left[x,\ x+\frac{1}{2},\ -z\right]$	[14,23]
15	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	[15,22]
16	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[16,17]

Table 13: Wyckoff site: 32m, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[4]
5	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$[y,\ -x,\ -z]$	[10]
11	$[-y,\ x,\ -z]$	[11]

continued ...

Table 13

No.	position	mapping
12	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[19]
20	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[20]
21	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[23]
24	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[25]
26	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[28]
29	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[29]
30	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[32]